

Operating Systems: Assignment-3

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Q 1. Write a Shell Script to find,

a) Largest of N numbers

Code:

```
#!/bin/bash

echo "Enter how many numbers:"
read n

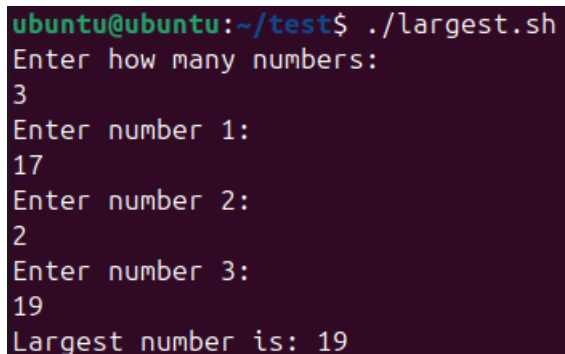
largest=0

for ((i=1; i<=n; i++))
do
    echo "Enter number $i:"
    read num

    if [ $i -eq 1 ] || [ $num -gt $largest ]; then
        largest=$num
    fi
done

echo "Largest number is: $largest"
```

Output:

A terminal window with a dark purple background. The prompt is 'ubuntu@ubuntu:~/test\$'. The user has run './largest.sh'. The script prompts 'Enter how many numbers:' and the user enters '3'. Then it prompts 'Enter number 1:', 'Enter number 2:', and 'Enter number 3:'. The user enters '17', '2', and '19' respectively. Finally, the script outputs 'Largest number is: 19'.

```
ubuntu@ubuntu:~/test$ ./largest.sh
Enter how many numbers:
3
Enter number 1:
17
Enter number 2:
2
Enter number 3:
19
Largest number is: 19
```

b) Swap two values

Code:

```
#!/bin/bash

echo "Enter first number:"
read a

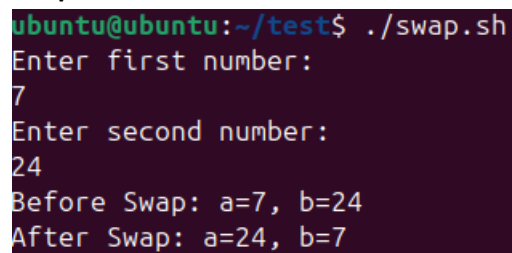
echo "Enter second number:"
read b

echo "Before Swap: a=$a, b=$b"

temp=$a
a=$b
b=$temp

echo "After Swap: a=$a, b=$b"
```

Output:



```
ubuntu@ubuntu:~/test$ ./swap.sh
Enter first number:
7
Enter second number:
24
Before Swap: a=7, b=24
After Swap: a=24, b=7
```

c) Odd or Even

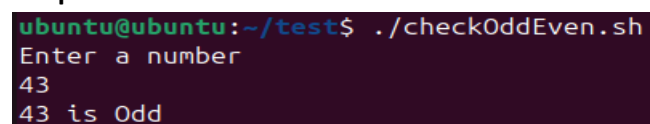
Code:

```
#!/bin/bash

echo "Enter a number:"
read num

if [ $((num % 2)) -eq 0 ]; then
    echo "$num is Even"
else
    echo "$num is Odd"
fi
```

Output:



```
ubuntu@ubuntu:~/test$ ./checkOddEven.sh
Enter a number
43
43 is Odd
```

d) Sum of three numbers

Code:

```
#!/bin/bash

echo "Enter first number:"
read a

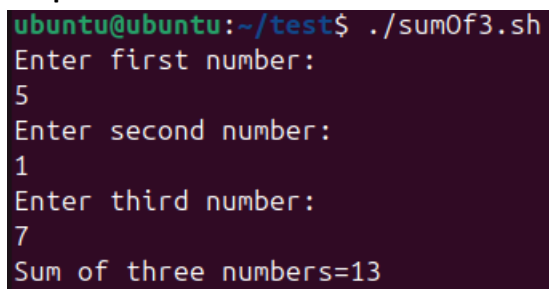
echo "Enter second number:"
read b

echo "Enter third number:"
read c

sum=$((a + b + c))

echo "Sum of three numbers = $sum"
```

Output:

A terminal window with a dark purple background. The prompt is 'ubuntu@ubuntu:~/test\$'. The user has run './sumOf3.sh'. The script prompts for three numbers: 'Enter first number:', 'Enter second number:', and 'Enter third number:'. The user has entered '5', '1', and '7' respectively. The final output is 'Sum of three numbers=13'.

```
ubuntu@ubuntu:~/test$ ./sumOf3.sh
Enter first number:
5
Enter second number:
1
Enter third number:
7
Sum of three numbers=13
```

Q 2. Write a shell script to check to see if the file "file_path" exists. If it does exist, display "file_path passwords are enabled." Next, check to see if you can write to the file. If you can, display "You have permissions to edit "file_path."" If you cannot, display "You do NOT have permissions to edit "file_path".

Code:

```
#!/bin/bash

echo "Enter the file path:"
read file_path

if [ -f "$file_path" ]; then
    echo "$file_path passwords are enabled."

    if [ -w "$file_path" ]; then
        echo "You have permissions to edit $file_path."
    else
        echo "You do NOT have permissions to edit $file_path."
    fi
else
    echo "$file_path does NOT exist."
fi
```

Output:

```
ubuntu@ubuntu:~/test$ ./file.sh
Enter the file path:
/home/ubuntu/test/addressbook.txt
/home/ubuntu/test/addressbook.txt passwords are enabled.
You do NOT have permissions to edit /home/ubuntu/test/addressbook.txt.
```

Q 3. Write a shell script that asks the user to type a number of words as input to shell script Store the input words by user into an array. Print how long each word is.

Code:

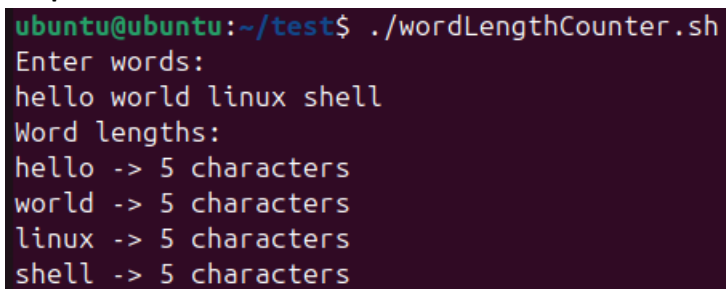
```
#!/bin/bash

echo "Enter words (separated by space):"
read -a words

echo "Word lengths:"

for word in "${words[@]}"
do
    length=${#word}
    echo "$word -> $length characters"
done
```

Output:

A terminal window with a dark purple background. The prompt is 'ubuntu@ubuntu:~/test\$'. The user has run './wordLengthCounter.sh'. The script prompts 'Enter words:' and the user has entered 'hello world linux shell'. The script then prompts 'Word lengths:' and outputs the character count for each word: 'hello -> 5 characters', 'world -> 5 characters', 'linux -> 5 characters', and 'shell -> 5 characters'.

```
ubuntu@ubuntu:~/test$ ./wordLengthCounter.sh
Enter words:
hello world linux shell
Word lengths:
hello -> 5 characters
world -> 5 characters
linux -> 5 characters
shell -> 5 characters
```

Q 4. Write a shell program to reverse the five-digit integer.

Code:

```
#!/bin/bash

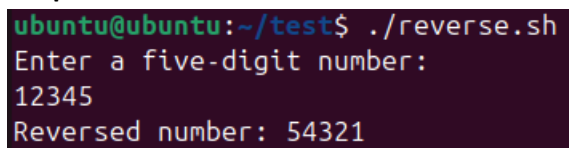
echo "Enter a five-digit number:"
read num

rev=0

while [ $num -gt 0 ]
do
    digit=$((num % 10))
    rev=$((rev * 10 + digit))
    num=$((num / 10))
done

echo "Reversed number: $rev"
```

Output:

A terminal window with a dark background. The prompt is 'ubuntu@ubuntu:~/test\$'. The user has entered './reverse.sh'. The script outputs 'Enter a five-digit number:', followed by the user input '12345'. The script then outputs 'Reversed number: 54321'.

```
ubuntu@ubuntu:~/test$ ./reverse.sh
Enter a five-digit number:
12345
Reversed number: 54321
```

Q 5. Write a shell program to search for a given number from the list of numbers using binary search.

Code:

```
#!/bin/bash

echo "Enter number of elements:"
read n

echo "Enter $n numbers in sorted order:"
for ((i=0; i<n; i++))
do
    read arr[$i]
done

echo "Enter number to search:"
read key

low=0
high=$((n-1))
found=0

while [ $low -le $high ]
do
    mid=$(( (low + high) / 2 ))

    if [ "${arr[$mid]}" -eq "$key" ]; then
        echo "Number found at position $((mid + 1))"
        found=1
        break
    elif [ "${arr[$mid]}" -lt "$key" ]; then
        low=$((mid + 1))
    else
        high=$((mid - 1))
    fi
done

if [ $found -eq 0 ]; then
    echo "Number not found in the list"
fi
```

Output:

```
ubuntu@ubuntu:~/test$ ./binarySearch.sh
Enter number of elements:
5
Enter 5 numbers in sorted order:
10
20
30
40
50
Enter number to search:
30
Number found at position 3
```